

A Finite Periodic Skeleton Disjointly Attached to an Aperiodic Minimal Subshift

Route-A structural certificate C149

Abstract

We attach tagged cycles of lengths 1, 2, 3, 5 by topological disjoint union to the aperiodic minimal Thue–Morse subshift. The resulting system has four primitive cycles and rational Artin–Mazur zeta $((1-z)(1-z^2)(1-z^3)(1-z^5))^{-1}$. This supplies a controlled finite periodic skeleton but destroys minimality. The construction is a declared attachment, not an almost-minimal system or an intrinsic creation of orbits inside the substitution language.

1 The disjoint system

Let X_{TM} be the two-sided subshift generated by $0 \mapsto 01$, $1 \mapsto 10$. It is nonempty and minimal and contains no periodic point. For $E = \{1, 2, 3, 5\}$, let C_ℓ be a tagged cycle of length ℓ and set

$$Y = X_{\text{TM}} \sqcup \bigsqcup_{\ell \in E} C_\ell.$$

The map is the shift on X_{TM} and cyclic successor on each C_ℓ . One iterate is the clock.

Aperiodicity of the infinite component. For completeness, let t_j be the parity of the binary digit sum of j . If $0 \leq r < 2^q$, binary concatenation gives $t_{j2^q+r} = t_j \text{ xor } t_r$. Fix a proposed period p and choose odd k larger than its binary length. For

$$d = p(2^k - 1) = (p - 1)2^k + (2^k - p),$$

the lower k bits complement the k -bit expansion of $p - 1$; hence $\text{popcount}(d) = k$ is odd. Thus $p \mid d$ but $t_d \neq t_0$. Let b be the binary length of d . Every interval of t of length 2^{b+1} contains a full block aligned at a multiple $j2^b$. Its symbols at offsets $0, d$ are $t_j \text{ xor } t_0$ and $t_j \text{ xor } t_d$, hence are opposite at a distance divisible by p . No such interval is p -periodic. A p -periodic point of X_{TM} would have a window of that length occurring in t , a contradiction. Therefore the infinite component contributes no positive-period fixed point.

2 All-period counts

The component C_ℓ contributes all ℓ of its points to $\text{Fix}(T^n)$ exactly when $\ell \mid n$. Therefore

$$\# \text{Fix}(T^n) = \sum_{\substack{\ell \in E \\ \ell \mid n}} \ell. \quad (1)$$

Each component is a single cycle of its displayed least period. Equivalently, if $P(n)$ counts exact-period points, then

$$P(n) = \sum_{d \mid n} \mu(n/d) \# \text{Fix}(T^d), \quad C(n) = P(n)/n,$$

so $C(1) = C(2) = C(3) = C(5) = 1$ and all other primitive counts vanish. With

$$\zeta_Y(z) = \exp \left(\sum_{n \geq 1} \# \text{Fix}(T^n) \frac{z^n}{n} \right),$$

we obtain

$$\begin{aligned} \log \zeta_Y(z) &= \sum_{\ell \in E} \sum_{q \geq 1} \frac{z^{\ell q}}{q} = - \sum_{\ell \in E} \log(1 - z^\ell), \\ \zeta_Y(z) &= \frac{1}{(1-z)(1-z^2)(1-z^3)(1-z^5)}. \end{aligned} \quad (2)$$

3 Minimality obstruction and validation

With the componentwise topology, a finite disjoint union of compact spaces is compact. Every tagged cycle is a proper nonempty closed invariant subset of Y , so Y is not minimal. The same proof applies to any nonempty finite union of periodic cycles disjointly attached to any nonempty aperiodic component. In particular, no almost-minimal conclusion is asserted.

A 60-period exact ledger and degree-30 zeta series agree with (1)–(2). The independent checker passes 395 assertions, SymPy passes 277 checks, replay is byte-identical, and 42 hostile cases are rejected.

The strict tuple is (A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL). The cycles are declared topological disjoint attachments, not points of X_{TM} . We claim no target divisor, arithmetic/local factor, target functional equation or counting law, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.